LLMs and Formal Methods: A Survey

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AI-assisted program synthesis is increasingly practical, but trustworthy use depends on formal specifications and mechanically checked correctness arguments. This survey is motivated by a verification-oriented agenda in which AI-generated code and proofs are grounded in a shared, human-reviewable intermediate specification, then validated by mechanised checking. The survey therefore focuses on research that helps formalise intent, generate verification-oriented code and proof artefacts, and connect informal requirements to proof-producing workflows. The emphasis is on systems with a direct formal-methods component rather than general coding assistance.

CCS Concepts: • **Software and its engineering** → **Software verification and validation**; • **Computing methodologies** → *Machine learning*; • **Theory of computation** → *Program verification.*

Additional Key Words and Phrases: formal methods, large language models, program verification, mechanised theorem proving, specification languages, neuro-symbolic methods

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1 TAXONOMY

The taxonomy is organised around the parts of a verification-oriented workflow most relevant to the motivating research direction: formalising intent, generating verification artefacts, synthesising or repairing proofs, and evaluating these systems on shared benchmarks. Adjacent specification-oriented tools are included only where they clarify interfaces between informal requirements, intermediate specifications, and mechanised verification.

Area	What sits here	Section
Surveys and roadmaps	High-level surveys, position papers, and roadmaps that frame the field and its major research directions.	§2

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Area	What sits here	Section
Specification languages and requirements	Natural language to logic/specification, intent formalisation, and auditable requirement capture.	§3
Verified code generation and proof synthesis for code	Verifier-in-the-loop generation of code, contracts, assertions, and proofs in deductive and contract-based program verification settings.	§4
Automated proof search, synthesis, and repair	Automation-oriented proof search, proof synthesis, proof repair, and verifier-guided reasoning that can support proof-producing verification workflows.	§5
Benchmarks and datasets	Shared evaluation sets for theorem proving, verifiable code generation, and repository-scale proof.	§6
Adjacent specification and AI-native coding systems	Spec-centric languages, AI-oriented implementation languages, and spec-driven development toolchains that are not themselves formal methods results but may shape how intent is structured before verification.	§7

2 SURVEYS AND ROADMAPS

- **Autoformalisation survey**. [Wen+25]
- **Survey on LLMs for software-requirement formalisation**. [BOM25]
- **Survey on deep learning for theorem proving**. [Li+24b]
- **Roadmap on LLMs and formal methods for trustworthy agents**. [Zha+24a]
- **Formal mathematical reasoning as an AI frontier**. [Yan+25c]
- **Two-way roadmap between requirements engineering and LLMs**. [FS25]

3 SPECIFICATION LANGUAGES AND REQUIREMENTS

- **VERIFAI**. Early work on traceability and verification of natural-language requirements, useful as background for NL-to-spec workflows. [24c]
- **Formal Specifications from Natural Language**. Direct NL-to-formal-spec framing paper. [Hah+22]
- **Autoformalization with Large Language Models**. Early LLM-based autoformalisation of informal mathematical text. [Wu+22]
- **Process-Driven Autoformalization in Lean 4**. Systematic pipeline for translating informal mathematical text into Lean 4 proofs with process supervision. [Lu+24]

- **Improving Autoformalization using Type Checking**. Shows that type-checking feedback significantly reduces autoformalization errors. [Poi+24]
- **nl2spec**. Interactive translation from unstructured requirements to temporal logic with editable sub-translations. [Cos+23]
- **NL2TL**. A general framework for mapping natural-language instructions to temporal logic across multiple domains. [Che+23]
- **SpecGen**. Automated generation of formal specifications from natural-language requirements. [Ma+25]
- **Specify What?** and **Seeking Specifications**. Neuro-symbolic work on deriving and refining specifications from informal descriptions. [GAJ24; GAJ25]
- **Doc2Spec**. A multi-agent approach that induces formal specifications from natural-language documents. [Xia+26]
- **Trustworthy Formal Natural Language Specifications**. A proof-assistant-native route to formalising natural-language specifications inside Lean. [GM23]
- **Microsoft intent formalisation agenda**. Industry framing that treats user-intent formalisation as a central reliability problem for AI-assisted coding. [25e]
- **nl2postcond** and follow-on evaluation work. Microsoft-led efforts on turning programmer intent into checkable postconditions and evaluating this route for verification-aware languages. [End+24; Lah24]
- **Supporting Software Formal Verification with Large Language Models: An Experimental Study**. Broader evidence on where LLM assistance helps in formal-specification and verification workflows. [Wan+25b]

4 VERIFIED CODE GENERATION AND PROOF SYNTHESIS FOR CODE

- **DafnySynth / dafny-annotator**. LLM-assisted synthesis of Dafny assertions, invariants, and proof-oriented annotations. [PLA24]
- **AI-assisted Dafny synthesis (Misu et al.)**. Systematic study of LLM-assisted synthesis of verified Dafny methods, evaluating multiple LLMs on generation and annotation tasks at FSE 2024. [Mis+24]
- **Laurel**. Uses LLMs to unblock automated verification by synthesising useful proof and specification steps. [Mug+25]
- **Verification with Ada/SPARK**. Evaluation of LLM-generated code in a SPARK-based verification setting. [CM25]
- **LLM code generation with formal specifications and reactive synthesis**. Couples code generation with formal specifications and synthesis constraints. [Mur+24b]
- **VeCoGen**. Formally verified C generation using LLMs plus a verifier-guided refinement loop. [SEN25]
- **Automatic generation of formal specification and verification annotations using LLMs and test oracles**. Annotation-generation workflow for verification tools. [Far+26]

- **Evaluating LLM-generated ACSL annotations**. Assessment of annotation quality for Frama-C-style verification workflows. [BOM26]
- **AlphaVerus**. Self-improving verified code generation for Verus/Rust. [25a]
- **RAG-Verus**. Repository-level proof synthesis for Verus with retrieval augmentation and repository context. [Zho+25]
- **Microsoft verus-proof-synthesis**. Umbrella repository for Microsoft's Verus verification automation stack. [25f]
- **AutoVerus**. Automated proof generation for Rust/Verus. [Yan+25a]
- **VeruSAGE**. Agent-based verification for repository-scale Rust systems. [Yan+25b]
- **VeriStruct**. Planner-based verification of Verus data-structure modules. [Sun+25]
- **Generating Safety-Critical Automotive C-programs using LLMs with Formal Verification**. Joint industry-academic evaluation in a safety-critical domain. [25c]
- **Clover**. Closed-loop verifiable code generation: LLM synthesises Dafny programs while a separate verifier loop checks and repairs annotations. [Sun+24]
- **SAFE**. Self-evolution approach to automated Rust/Verus proof generation, using iterative fine-tuning on successful proof traces. [Che+24]
- **Enchanting**. Uses LLMs combined with static analysis and program verification to synthesise formal specifications from code. [Wen+24]
- **Rango**. Adaptive retrieval-augmented proof generation for Coq/Rocq: retrieves similar verified proofs to guide LLM-based proof synthesis. [Tho+24]
- **Neural Synthesis for SMT-Assisted Proof-Oriented Programming**. Fine-tunes LLMs to generate F* proof obligations dischargeable by an SMT solver, targeting proof-oriented language verification. [Cha+24]
- **Lemur**. Integrates LLMs into automated program verification pipelines (Dafny/Frama-C), combining deductive verification with LLM-generated invariants and annotations. [WBN23]
- **Finding Inductive Loop Invariants using LLMs**. Uses LLMs to synthesise candidate loop invariants, verified by Frama-C/Boogie, with iterative feedback. [Kam+23]
- **Evaluating LLMs for VeriFast specifications**. Evaluates the ability of LLMs to generate verifiable memory-safety specifications in VeriFast for C programs. [Reg+24]
- **Can LLMs Enable Verification in Mainstream Programming?** Empirical study of LLM-assisted deductive verification across mainstream languages (Java/JML, C/Frama-C), assessing annotation correctness. [She+25]

5 AUTOMATED PROOF SEARCH, SYNTHESIS, AND REPAIR

- **GPT-f**. Early neural theorem-proving baseline for proof search guidance. [PS20]
- **ProofNet**. Autoformalisation and formal proving for undergraduate-level mathematics. [Aze+23b]
- **Baldur**. Whole-proof generation and repair with large language models. [Fir+23]

- **LeanDojo / ReProver**. Open tooling and datasets for Lean-based proving with retrieval-augmented models. [Yan+23]
- **COPRA**. In-context agentic theorem proving for interactive assistants. [TWC23]
- **DeepSeek-Prover**. Open Lean prover line with synthetic-data training; the public repository tracks released models. [Xin+24; 25b]
- **SubgoalXL**. Subgoal-based expert learning for theorem proving. [Zha+24b]
- **LEGO-Prover**. Neural theorem proving with growing libraries. [Wan+23]
- **APOLLO**. Lean+LLM collaboration driven by compiler-guided repair. [OFY25]
- **HybridProver**. Combines theorem proving with LLM-driven proof synthesis and repair. [Hu+25]
- **A Minimalist Proof Language for Neural Theorem Proving over Isabelle/HOL**. Proof-language design for neural theorem proving over Isabelle/HOL. [Xu+25a]
- **Agentic Program Verification / AutoRocq**. Agentic Rocq/Coq verification via iterative prover feedback. [Tu+25]
- **AutoReal on seL4**. LLM-driven theorem proving in an industrial-scale verification setting. [Zha+26a]
- **Vibe Coding an LLM-powered Theorem Prover / Isabellm**. Lightweight Isabelle/HOL proving with both a paper and an open implementation. [Hou26; 26b]
- **LeanNavigator**. Large-scale generation of additional Lean theorems and proofs from Mathlib4. [YG25]
- **AlphaProof programme**. Google DeepMind's formal mathematical reasoning effort, framed publicly through reinforcement learning and formal proof search. [24a; 25g]
- **Harmonic / Aristotle**. Industry effort on olympiad-level automated theorem proving and formal mathematical reasoning. [Ach+25]
- **formal-conjectures**. Repository of Lean-formalised conjectures intended to support formal reasoning research. [24b]
- **TheoremLlama**. Fine-tunes general-purpose LLMs on Lean 4 proof data to create Lean-expert models. [Wan+24b]
- **FVEL**. Interactive formal verification environment coupling LLMs with Isabelle/HOL for software property proving. [Lin+24b]
- **Selene**. Pioneer work on automated proof generation for software verification, bridging LLMs with Lean for repository-scale proofs. [ZLD24]
- **HyperTree Proof Search**. Tree-search algorithm for neural theorem proving in Lean and Metamath, with online training via MCTS-style exploration. [Lam+22]
- **Lean-STaR**. Trains LLMs to interleave informal thinking steps with Lean proof steps, improving proof generation quality. [Lin+24a]
- **REFACTOR**. Learns to identify and extract reusable lemmas from existing Lean proofs to grow the library. [Zho+24]

- **Graph2Tac**. Graph neural network approach to premise selection and tactic prediction for Lean/Coq. [Rut+24]
- **Formal Mathematics Statement Curriculum Learning**. Uses expert iteration and curriculum learning to train LLMs for formal mathematics in Lean and Metamath. [Pol+22]
- **Llemma**. Open language model specialised for mathematics, pre-trained on formal and informal mathematical text; widely used as a backbone for theorem-proving models. [Aze+23a]
- **Goedel-Prover** and **Goedel-Prover-V2**. Frontier open-source Lean provers trained on large synthetic datasets, with V2 adding scaffolded data synthesis and self-correction. [Lin+25b; Lin+25a]
- **DeepSeek-Prover-V2**. Advances Lean theorem proving with reinforcement learning over subgoal decomposition trees. [Ren+25]
- **Kimina-Prover**. Large formal reasoning model trained with RL, achieving strong miniF2F results. [Wan+25a]
- **AlphaGeometry**. Neurosymbolic system that solves olympiad geometry problems at silver-medal level using a language model paired with a symbolic geometry engine. [Tri+24]
- **Autoformalizing Euclidean Geometry**. Demonstrates autoformalization of Euclidean geometry into Lean, translating informal geometric statements and proofs. [Mur+24a]
- **LINC** and **Logic-LM** and **SatLM**. Neurosymbolic approaches that combine LLMs with first-order logic provers, symbolic solvers, and SAT/SMT reasoners respectively to achieve faithful formal reasoning. [Ola+23; Pan+23; Ye+23]
- **Proof Artifact Co-Training (PACT)**. Jointly trains a language model on proof artifacts (tactic states, intermediate goals) alongside informal text, improving formal proof generation across interactive proof assistants. [Han+22]
- **Lyra**. Orchestrates dual correction in automated theorem proving: an LLM generates proof drafts while a separate refinement loop corrects both the informal reasoning and formal proof steps. [Zhe+23]
- **Subgoal-based Demonstration Learning**. ICML 2024 approach that decomposes formal proofs into subgoals and uses demonstration learning to train LLMs on structured subgoal sequences for Lean theorem proving. [ZLK24]
- **HunyuanProver**. Scalable data synthesis framework combined with guided tree search for automated Lean theorem proving, achieving strong miniF2F results. [Li+24a]
- **Proving Theorems Recursively**. Decomposes complex theorems into simpler subgoals recursively, using LLMs to prove each subgoal independently and assemble the full proof. [Wan+24a]
- **BFS-Prover**. Scalable best-first tree search for LLM-based automatic theorem proving in Lean, with online value estimation guiding proof search. [Xin+25]

- **InternLM2.5-StepProver**. Advances automated Lean theorem proving via expert iteration on large-scale problem sets, training step-level models with iterative data synthesis. [Wu+24]

6 BENCHMARKS AND DATASETS

- **miniF2F**, **PUTNAMBENCH**, and **FormalMATH**. Benchmarks for formal mathematical reasoning and theorem proving across systems and domains. [ZHP22; Tso+24; Yu+25]
- **FIMO**. Challenge dataset for automated theorem proving. [Liu+23]
- **CLEVER** and **VERINA**. Benchmarks targeting verifier-in-the-loop code generation and proof-oriented programming tasks. [Tha+25; Ye+25]
- **A benchmark for vericoding**, **AlgoVeri**, and **CASP**. Evaluation sets for formally verified program synthesis and code-level proof generation. [Bur+25; Zha+26b; Her+25]
- **DafnyBench** and **DafnyCOMP**. Benchmarks and competitions for LLM-assisted Dafny verification workflows. [Lou+25; Xu+25b]
- **VeriSoftBench**. Repository-scale Lean benchmark focused on software-verification obligations rather than pure mathematics. [Xin+26]
- **miniCodeProps**. Minimal Lean 4 benchmark for proving properties of small programs, useful for evaluating code-verification LLMs. [LW24]
- **Proving the Coding Interview**. Lean 4 benchmark built from interview-style algorithmic problems requiring formal verification proofs. [DM25]
- **Lean Workbook**. Large-scale dataset of Lean 4 problem statements collected from competition mathematics, intended to support training and evaluation of formal proving systems. [Yin+24]
- **Herald**. Benchmark for human-in-the-loop formal reasoning evaluation, measuring LLM ability to assist human provers on interactive theorem-proving tasks. [Gao+24]
- **miniCTX**. Context-dependent theorem proving benchmark for Lean that requires reasoning about file-level and project-level context beyond local tactic state. [HZW24]

7 ADJACENT SPECIFICATION AND AI-NATIVE CODING SYSTEMS

- **Mog**. An AI-agent-oriented embedded language designed to be written by LLMs, with an explicit capability model and a compact language specification. [26c]
- **CodeSpeak**. A spec-centric programming language and toolchain that generates conventional code from maintained specifications using LLMs. [26a]
- **Spec Kit**. GitHub's open-source toolkit for spec-driven development with AI agents, centred on turning specifications into implementation plans and executable workflows. [25d]

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